

Two True Stories

Team Members: _____

Team Role (circle one): THE REGISTRAR THE LEARNING CENTER

The Situation

Last year both sections of Elements of Data Science offered an optional review session before Quiz 2, and the Provost's office wants to know whether it was worth the money. Every student who took Quiz 2 is counted below. Each had already taken Quiz 1; those who scored below 70 on Quiz 1 are listed as **Struggling**, the rest as **Strong**.

Section	Quiz 1 group	Review session	Students	Passed Quiz 2
A	Strong	Attended	12	11
B	Strong	Attended	8	7
A	Strong	Did not attend	44	37
B	Strong	Did not attend	36	31
A	Struggling	Attended	28	15
B	Struggling	Attended	32	15
A	Struggling	Did not attend	9	4
B	Struggling	Did not attend	11	4

Your team has a client, and your client has already decided what they believe. Your job is to **find it in this table**.

THE REGISTRAR wants to cancel the sessions: “*Students who attend review sessions do worse.*”

THE LEARNING CENTER wants to expand them: “*Review sessions help students.*”

No row may be dropped, no number invented, no student ignored. The only thing you may choose is **how to group**.

Part 1: Build Your Table (5 minutes)

Decide which rows belong together, combine them, and compute a pass rate for each group. Write the table you would hand your client.

Group	Students	Passed	Pass rate

Part 2: The Other Team (4 minutes)

When the other side reads their headline and their table: did they cheat? Point to the step where they went wrong — or explain why you cannot.

Discussion Questions

Both tables are correct. Every number in both of them can be recomputed from the data you were given, and neither team dropped a single student.

If two contradictory conclusions can both be honestly derived from one dataset, what decides which one is right? Notice that the answer is not in the data.

Both teams combined Section A with Section B and nothing happened. Only one team combined Strong with Struggling — and the answer reversed.

Both moves were the same operation. What is different about the two columns? State the property a column must have before collapsing it can change your conclusion.

Look again at the counts. Sixty of the eighty students who attended were already struggling; only twenty of the hundred who skipped it were.

Why did that happen, and what does it tell you about what the overall comparison is actually comparing?

Suppose Quiz 1 scores had never been recorded.

Would anyone have been able to detect the problem? A grouped table always looks like a summary of the data. When is it better described as an argument about the data?